

UNITED TECHNICAL COLLEGE

Level: Bachelor

Semester: Fall

Year : 2025

Programme: BE

Full Marks : 50

Course: Network and Cyber Security

Pass Marks : 25

Semester: 7th

Time : 1.5hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt any three questions from 1-6. Q.N.7 is Compulsory

- 1 a. ✓ Define computer security. Explain different types of passive and active attacks with appropriate examples. 7
b. ✓ Describe the CIA triad (Confidentiality, Integrity, and Availability) and the role of firewalls in maintaining network boundaries. 8
- 2 a. ✓ Differentiate between stream ciphers and block ciphers. Briefly explain the structure of the Feistel Cipher. 7
b. ✓ Using the Playfair cipher algorithm, encrypt the plaintext "CRYPTOGRAPHY" using the keyword "SYMMETRIC". 8
- 3 a. ✓ Explain the RSA algorithm and discuss the significance of the Prime Factorization Problem in its security. 7
b. ✓ Describe the Diffie-Hellman Key Exchange protocol. How does the Discrete Logarithm Problem provide its mathematical foundation? 8
- 4 a. Describe the logic of the SHA-512 algorithm and explain the purpose of the padding and length-appending steps. 8
b. What is a Message Authentication Code (MAC)? Explain how HMAC provides a more secure authentication mechanism than simple prefix/suffix hashing. 7
- 5 a. Explain the Secure Socket Layer (SSL) architecture. Describe the four phases of the Handshake Protocol. 7
b. Distinguish between IPsec Transport Mode and Tunnel Mode. What are the primary security services provided by the Encapsulating Security Payload (ESP)? 8
- 6 a. What is the OWASP Top 10? Explain the concept of Server-Side Request Forgery (SSRF) and how to prevent it. 7

- b. Discuss the importance of Data Protection in storage and transit. What are Cryptographic Failures in web application security? 8

7 Write short notes on(Attempt Any One): 5

- a. DMZ (Demilitarized Zone)
- b. Kerberos Authentication Service
- c. Digital Certificates and Certificate Authorities